

Harini V

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PROFILE

M.Tech Software Engineering student with experience in developing AI-powered applications and workflow automation systems using Python, FastAPI, ReactJS, and SQL databases. Skilled in LLMs, RAG, Prompt Engineering, Generative AI, AI workflow evaluation, debugging, and performance monitoring. Experienced in designing and optimizing prompts, evaluating AI-generated outputs, and building intelligent AI-driven solutions. Strong foundation in software engineering, problem-solving, and analytical thinking, with a keen interest in AI agents, automation, and real-world LLM applications.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

AI & Generative AI: LLMs (ChatGPT, Gemini, Claude), RAG, Prompt Engineering, NLP, OpenAI APIs, AI Workflow Evaluation, Model Evaluation, Zero-shot, Few-shot, Chain-of-Thought, Role-based Prompting, Structured Output Generation

Frameworks & Libraries: FastAPI, ReactJS, Scikit-learn, Pandas, NumPy

Databases & Tools: MySQL, PostgreSQL, MongoDB, Git, GitHub, Docker, AWS (Fundamentals), Linux, VS Code, Redis

Testing & Debugging: Test Planning, Functional Testing, Regression Testing, Debugging, Log Analysis, Root Cause Analysis

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, REST APIs, JSON

Soft Skills: Problem Solving, Analytical Thinking, Team Collaboration, Communication

PROJECTS

AI-Assisted Engineering Test Platform | [GitHub](#) | *Python, FastAPI, PostgreSQL, Redis, Celery, Streamlit, Gemini AI, Docker*

- Developed an AI-powered engineering test platform that automates requirement analysis, test plan generation, test case creation, artifact evaluation, and root-cause analysis using Google Gemini and FastAPI microservices.
- Built a scalable backend with JWT authentication, PostgreSQL, Redis, and Celery for asynchronous workflow execution, audit logging, result tracking, and real-time AI job management.
- Designed an interactive Streamlit dashboard and containerized deployment using Docker Compose, enabling end-to-end test management, analytics visualization, reporting, and API-driven collaboration.

Prompt Evaluation Studio | *Python, FastAPI, Streamlit, OpenAI API, Gemini API*

- Built a multi-model prompt engineering platform to design, test, and compare prompts across GPT and Gemini models using zero-shot, few-shot, role-based, and chain-of-thought prompting techniques.
- Developed automated evaluation pipelines to measure response accuracy, latency, token consumption, cost efficiency, hallucination rates, and structured JSON output compliance.
- Created an interactive Streamlit dashboard for model comparison, prompt performance analytics, AI output evaluation, and prompt optimization.

AI-Powered Multilingual Crop Disease Assistant | *Python, Gemini API, NLP, RAG*

- Developed an AI assistant integrating a knowledge base with LLMs for crop disease diagnosis and contextual recommendation generation.
- Implemented Retrieval-Augmented Generation (RAG) and LLM-based response generation to provide accurate, context-aware recommendations for agricultural queries.
- Enabled multilingual text and voice interactions using NLP, translation, and speech technologies to improve accessibility.

PATENTS

Method and System for Nutrient Deficiency Classification in Rose Leaves Using Image Analysis — Developed an AI-based computer vision system for automated detection and classification of nutrient deficiencies in rose leaves using image preprocessing and feature analysis for early plant health diagnosis. (*Published, 2026*)

Telemetry-Driven Dynamic Logic-Switching System for Autonomous Agent Orchestration — Designed a fault-tolerant architecture that dynamically switches between AI reasoning engines and deterministic logic engines while preserving execution state for continuous task completion. (*Published, 2026*)

EDUCATION

Vellore Institute of Technology, Vellore
M.Tech Integrated Software Engineering — **CGPA: 8.66**

09/2022 – 07/2027 (Expected)

Orange International Matric Hr Sec School, Arni
Higher Secondary Education (Class XII) — **90.2%**

06/2021 – 05/2022

CERTIFICATIONS

Data Science Training Certification (VIT University)
Data Analytics Master Class (NoviTech R&D Pvt. Ltd.)
MEAN Stack (Guvi HCL)